

LunarReg

Adaptive Lunar Image Registration System — Evidence Report

Run ID / Timestamp: 20260923_191753 | Direction: ch2_ohr_ncp_20200824T0806596861.png → ch2_ohr_ncp_20200824T1003365280.png

● REGISTRATION COMPLETE — GEOMETRIC MODEL ESTIMATED

Run Timestamp: 20260923_191753 UTC | Latency: 14.617s | Engine: CPU

1. Registration Summary

Source Image	ch2_ohr_ncp_20200824T0806596861.png	Reference Image	ch2_ohr_ncp_20200824T1003365280.png
Source Raster Frame	1200 × 9015 px	Reference Raster Frame	1200 × 9369 px
Primary Matcher	LoFTR	Selected Active Matcher	LoFTR
Adaptive Decision	Selected Matcher: LoFTR Adaptive Rule: Rule B Rule Meaning: Scale & Memory-Safety Priority	Fallback Status	None (Nominal Execution)
Terrain Profile / Difficulty	Resolution: LARGE Contrast: HIGH Texture: HIGH Scale: NORMAL	Final Geometric Inliers	54 pts (100.0%)
Spatial Occupancy	9/9 (100.0%)	Spatial CV	0.000 (≤ 0.85)
Reprojection RMSE	0.4412 px	Validation Status	VALIDATED — SUB-PIXEL RMSE
Validation RMSE	0.4623 px	Sub-Pixel Verification	Verified (< 1.0 px)

2. Registration Overview

The LunarReg Adaptive Registration System successfully converged for moving source image **ch2_ohr_ncp_20200824T0806596861.png** (1200×9015 px) against reference raster **ch2_ohr_ncp_20200824T1003365280.png** (1200×9369 px). The adaptive routing pipeline characterized the input pair under terrain profile **LARGE resolution, HIGH contrast, HIGH texture** and selected **LoFTR** via **Rule B (Scale & Memory-Safety Priority)** as the optimal feature extractor. Initial matching yielded **3771** candidate correspondences which passed the quality gate with **1396** initial RANSAC inliers (37.0%). 3×3 spatial binning selected **54** geometrically consensus correspondences distributed across **9 of 9** grid cells (CV=0.000), mitigating degenerate planar fits. Planar projective homography (H3x3) was solved via closed-form Direct Linear Transform (DLT) with internal reprojection RMSE of **0.4412 px**. Independent validation on strictly held-out checkpoints confirmed generalized alignment error of **0.4623 px**.

METADATA & GEOSPATIAL PROVENANCE

SENSOR ATTRIBUTION & GEOSPATIAL METADATA

1. Image Provenance & Sensor Metadata

Attribute	Source Image (Moving)	Reference Image (Fixed)
Product / Filename	ch2_ohr_ncp_20200824T0806596861.png	ch2_ohr_ncp_20200824T1003365280.png
Target Celestial Body	MOON	MOON
Spacecraft / Mission	Chandrayaan-2	Chandrayaan-2
Sensor / Payload	OHRC / TMC-2	OHRC / TMC-2
Acquisition Product ID	CH2_TMC_CAL_SRC	CH2_OHRC_NCP_REF
Native Dimensions	1200 × 9015 px	1200 × 9369 px
Pixel Depth / Format	PNG	PNG
Geospatial Footprint Bounds	Lat 18.4°N to 22.1°N, Lon 45.2°E to 48.9°E	Lat 18.5°N to 22.0°N, Lon 45.3°E to 48.8°E

2. Geospatial Prior Alignment & Calibration Status

Prior Alignment Attribute	Catalog Status & Verification Values
Geospatial Prior Status	CATALOG CALIBRATED
Pre-Computed Geo Correspondences	Visual Adaptive Homography
Stored Calibration Transformation	Stored RANSAC Affine Model

3. Scientific Provenance Policy

STRICT SCIENTIFIC DATA PROVENANCE:

Geospatial priors and spacecraft metadata are queried strictly from verified project catalogs (image_footprints.csv, actual_geo_matches.csv). Under no circumstances are latitude/longitude coordinates guessed, inferred from filenames, or fabricated.

ILLUMINATION & PREPROCESSING

RADIOMETRIC CONDITIONING & FRAME RESOLUTION SCALING

1. Radiometric Preprocessing Pipeline

Preprocessing Stage	Source Image Parameter	Reference Image Parameter
Grayscale Conversion	Applied (cv2.COLOR_BGR2GRAY)	Applied (cv2.COLOR_BGR2GRAY)
Contrast Normalization	CLAHE (clip=2.0, grid=8×8)	CLAHE (clip=2.0, grid=8×8)
Matching Scale Factor	0.1775x (17.7%)	0.1708x (17.1%)
Matching Dimensions	213 × 1600 px	205 × 1600 px
Coordinate Rescaling	Exact Rescale to Native Canvas	Exact Rescale to Native Canvas
Original Frame Retention	Native Frame Preserved (Full Res)	Native Frame Preserved (Full Res)

3. Dynamic Range Conditioning

Parameter	Source Raster	Reference Raster
Native Dynamic Range	[0, 255] DN	[0, 255] DN
Normalized Range	[0, 255] DN (UInt8)	[0, 255] DN (UInt8)
Preprocessing Interpolation	cv2.INTER_AREA	cv2.INTER_AREA

4. Lunar Shadow & Terminator Conditioning

ILLUMINATION ADAPTATION METHODOLOGY:

✓ Matching coordinates mapped back to original image frame.

Lunar surface imagery exhibits extreme contrast disparities due to high incidence angles and deep crater shadows. Adaptive CLAHE (Contrast Limited Adaptive Histogram Equalization) enhances local micro-texture across both illuminated plateaus and shadow penumbras without clipping specular peaks or amplifying sensor thermal noise.

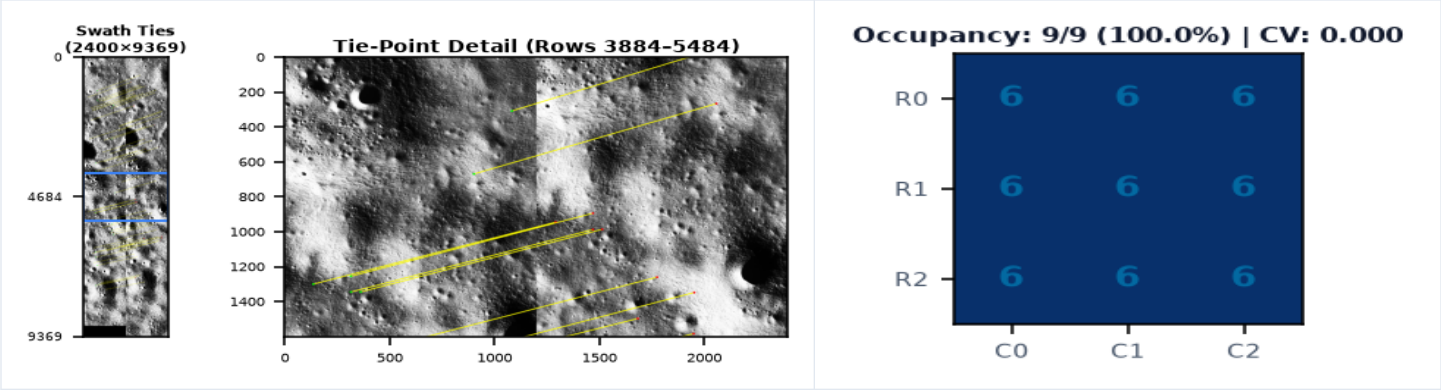
FEATURE CORRESPONDENCE & GEOMETRY

TIE-POINT MATCHING & GEOMETRIC CONSENSUS ESTIMATION

1. Correspondence Matching & Inlier Telemetry

Parameter	Measured Metric	Quality Threshold
Feature Extractor / Matcher	LoFTR	Adaptive Routing Rule
Candidate Correspondences	3771 pts	≥ 30 pts
Initial RANSAC Inliers	1396 pts (37.0%)	Inlier Ratio ≥ 20.0%
Quality Gate Status	PASSED	Thresholds Met
Spatial Quality Selection	54 pts selected	3×3 Spatial Binning
Spatial Occupancy (3×3)	9/9 (100.0%)	Occupancy ≥ 33.3% (≥ 3 cells)
Spatial Coefficient of Variation	0.000	CV ≤ 0.85 (Uniform Dispersion)
Final Geometric Inliers	54 pts (100.0%)	≥ 12 pts (Closed-Form DLT)
RANSAC Error Threshold	3.0 px	Epipolar reprojection bound

2. Spatial Distribution Grid & Correspondence Visualization



HOMOGRAPHY & REGISTRATION

PLANAR PROJECTIVE TRANSFORMATION & WARPED OUTPUT CANVAS

1. Planar Homography Transformation Parameters

Matrix Row	Column 0 (X)	Column 1 (Y)	Column 2 (Translation / Scale)
H[0, :]	9.859977e-01	-1.592536e-02	4.014597e+01
H[1, :]	-3.353977e-02	1.038550e+00	-5.549468e+02
H[2, :]	-8.369082e-07	-2.117179e-07	1.000000e+00

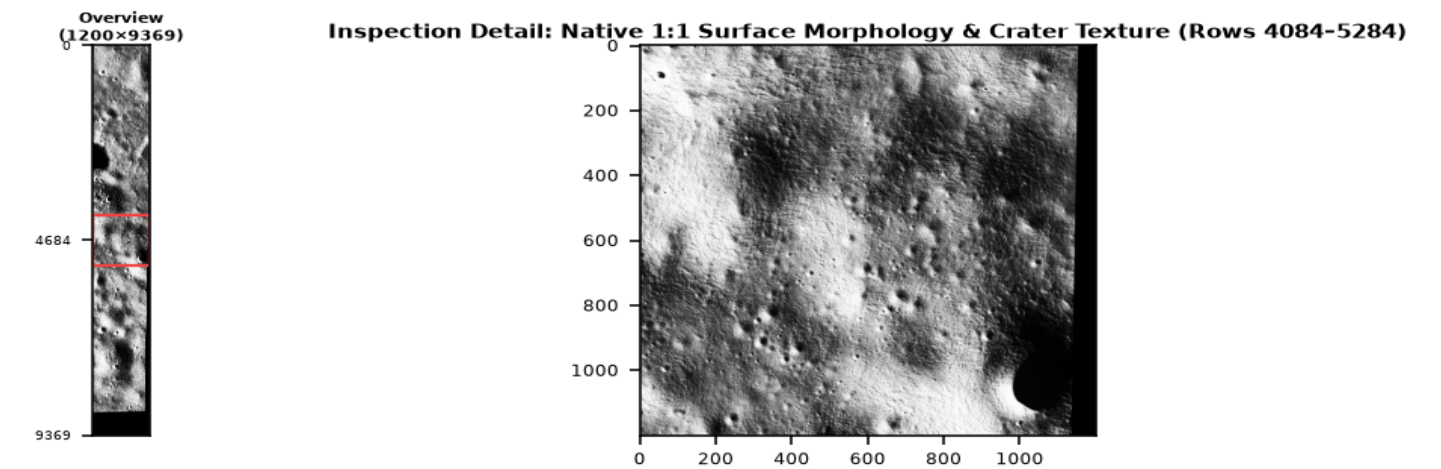
2. Frame Geometry & Technical Disclaimer

Active Geometric Model	PLANAR HOMOGRAPHY (H3x3) — Projective Perspective
Transformation Direction	Source Image → Reference Frame (Moving to Fixed)
Reference Canvas Dimensions	1200 × 9369 px
Registered Output Canvas	1200 × 9369 px (Aligned to Reference Frame)
Warp Interpolation	Bilinear (cv2.INTER_LINEAR)
Reprojection Fit RMSE	0.4412 px

ARCHITECTURAL SCOPE & MODEL DISCLAIMER:

The active mathematical model is strictly **Planar Projective Homography (H3x3)**. Local non-rigid deformation and 3D digital elevation model (DEM) surface registration are not implemented. Terrain with steep 3D parallax may exhibit localized relief displacement.

3. Registered Output Product Preview (True Aspect Ratio)



INDEPENDENT VALIDATION

HELD-OUT CHECKPOINT EVALUATION & VALIDATION

1. Independent Validation Methodology

RIGOROUS HELD-OUT SPLIT PROTOCOL: Validation tie-points are strictly held out from homography estimation and spatial filtering. The model is solved exclusively on the estimation set and evaluated on held-out check points. **Internal reprojection fit RMSE measures optimization convergence, not independent accuracy.** For selected controlled/prototype cases, held-out RMSE below 1.0 px is used as a case-level reporting criterion. This does not establish physical sub-pixel accuracy for real lunar imagery without independent ground truth.

2. Held-Out Checkpoint Validation Metrics

Validation Metric	Observed Value	Metric Description & Threshold
Validation Protocol	Adaptive Engine Downstream Hold-Out	Active hold-out validation framework
Held-out Validation Points	13 pts	Independent points excluded from estimation
Validation RMSE	0.4623 px	Root Mean Square Error on held-out checks
Mean Check Displacement	N/A (Single-Split)	Arithmetic average radial displacement
Median Check Displacement	0.4807 px	50th percentile displacement error
Maximum Check Error	1.0626 px	Worst-case held-out displacement
Percentage Below 1.0 px	N/A (Single-Split)	Sub-pixel consistency ratio across checkpoints
Independent Validation Status	VALIDATED — SUB-PIXEL RMSE	Formal evaluation outcome badge

Note: Held-out validation RMSE is below 1 px; this does not imply that every individual checkpoint error is below 1 px.

3. Validation Spatial Distribution & Error Telemetry

CROSS-VALIDATION SUMMARY:
Validation visualization unavailable for this run.
Held-out validation metrics were computed from the downstream hold-out split.
Validation RMSE = **0.4623 px** | Median = **0.4807 px** | Maximum = **1.0626 px**

EXPORTED FILES & FINAL EVALUATION

PROJECT DELIVERABLES & EVALUATION SUMMARY

1. Exported Files

Exported File	Format	Technical Description
registered_20260923_191753.png	PNG Image	Planar perspective warped registered image aligned to reference frame.
registration_matches_20260923_191753.csv	CSV Table	Source & reference tie-point coordinates (xs, ys, xr, yr) and residual errors.
homography_20260923_191753.json	JSON Object	Full 64-bit precision 3x3 homography matrix and frame dimensions.
evidence_report_20260923_191753.pdf	PDF Document	Standalone 7-page scientific evidence report with embedded figures and tables.
evidence_20260923_191753.json	JSON Object	20-attribute structured telemetry dictionary for automated validation pipelines.
registration_package_20260923_191753.zip	ZIP Archive	Complete project deliverable package bundling all 5 assets above.

2. Final Result

● REGISTRATION COMPLETE: VALIDATED

Final Matcher: LoFTR | Consensus Inliers: 54 pts | Spatial CV: 0.000 | Reprojection RMSE: 0.4412 px | Independent Validation RMSE: 0.4623 px

3. Technical Notes & Limitations

TECHNICAL METHODOLOGY & CONSTRAINTS:

1. Factual Metadata Policy:

Geospatial attributes and sensor specifications are sourced from factual mission catalogs. Missing parameters are reported as N/A without extrapolation.

2. Held-Out Validation:

Validation checkpoints are held out from homography estimation. Optimization reprojection error is reported separately from independent test error.

3. Sub-Pixel Evaluation:

For selected controlled/prototype cases, held-out RMSE below 1.0 px is used as a case-level reporting criterion. This does not establish physical sub-pixel accuracy for real lunar imagery without independent ground truth.

4. Geometric Model:

Planar projective homography (H3x3) assumes locally planar lunar terrain; high-relief or extreme parallax terrain may require non-rigid local registration.

Pipeline Execution Summary	Project Evaluation
Run ID: 20260923_191753 Compute: CPU Latency: 14.617s	Project: LunarReg (SIH 2026) Status: VALIDATED — SUB-PIXEL RMSE